Gripp

딥러닝 영상 분석을 활용한
클라이밍 경쟁 서비스

Team #5: 못올라가요 (나종우, 장원범, 조준오)

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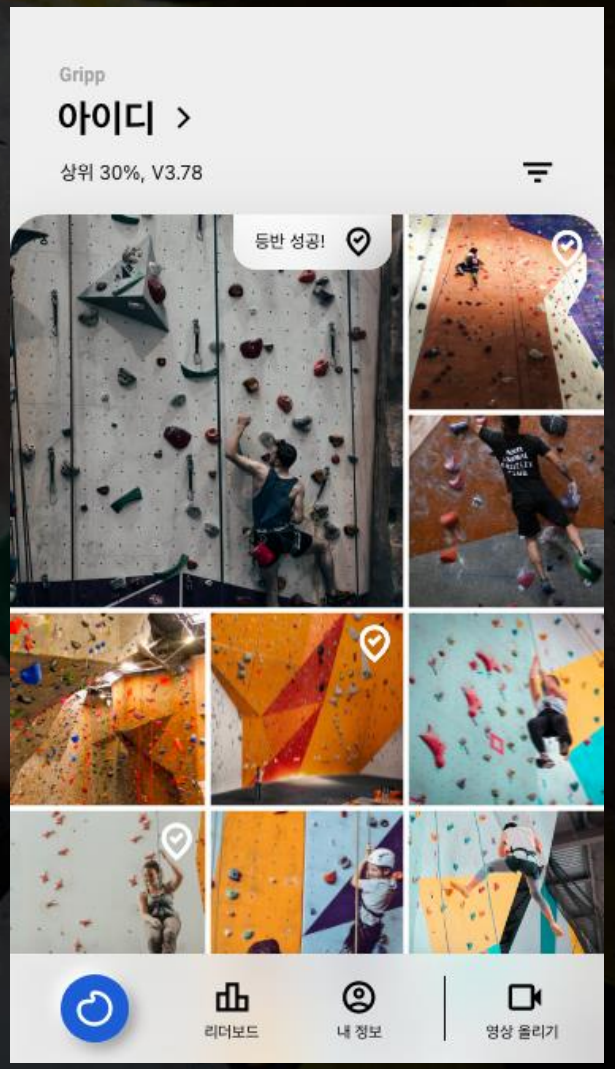
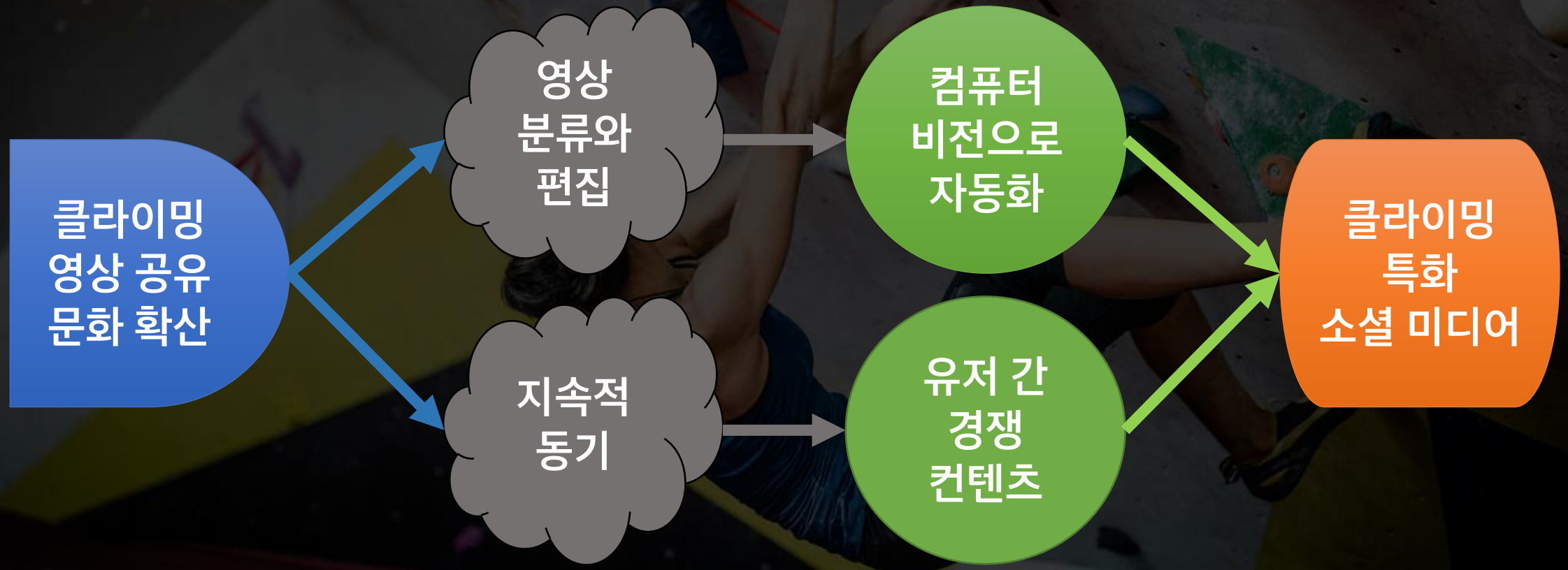


Gripp

Gripp = Grip + People



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개발 진행사항 - 컴퓨터비전

- 1차 데이터셋 제작
- Baseline Model 학습
- API 서버 제작 후 배포
- Detection, Pose Estimation Module 제작

PyTorch
MediaPipe
GitHub Actions
django
CVAT
Weights & Biases
MySQL

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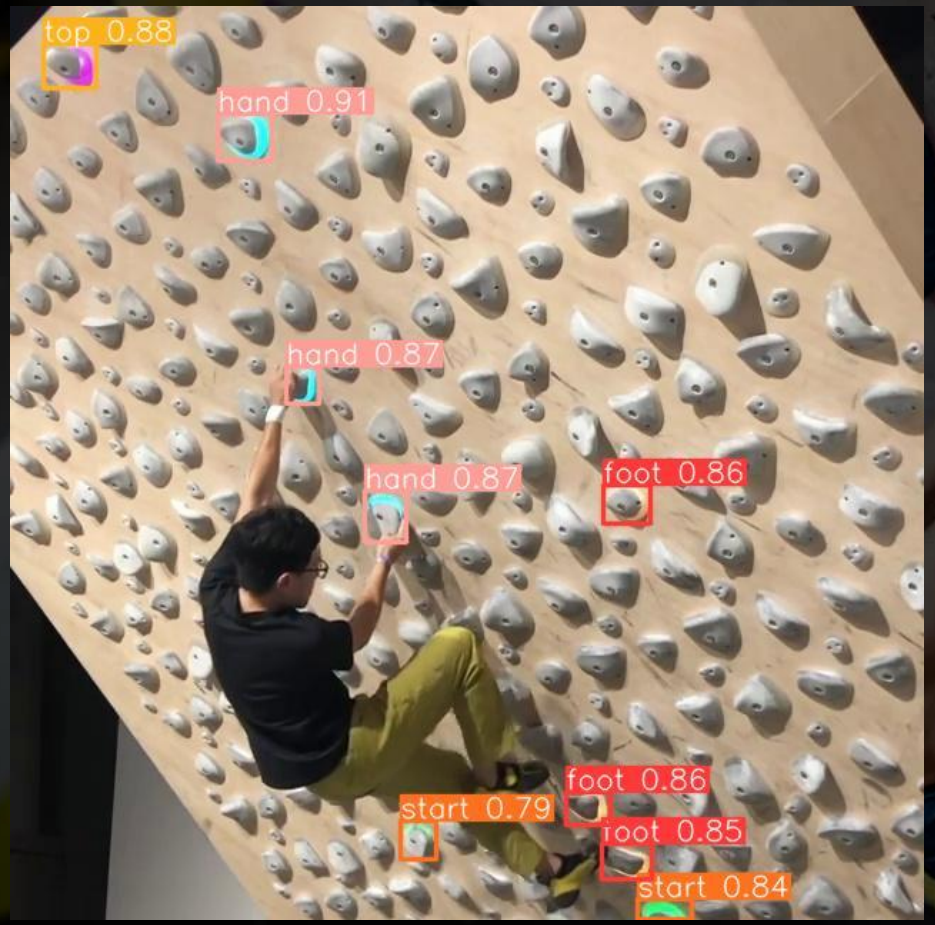
개발 진행사항 - 컴퓨터비전



Object Detection (YOLOv5)

Pose Estimation (mediapipe)

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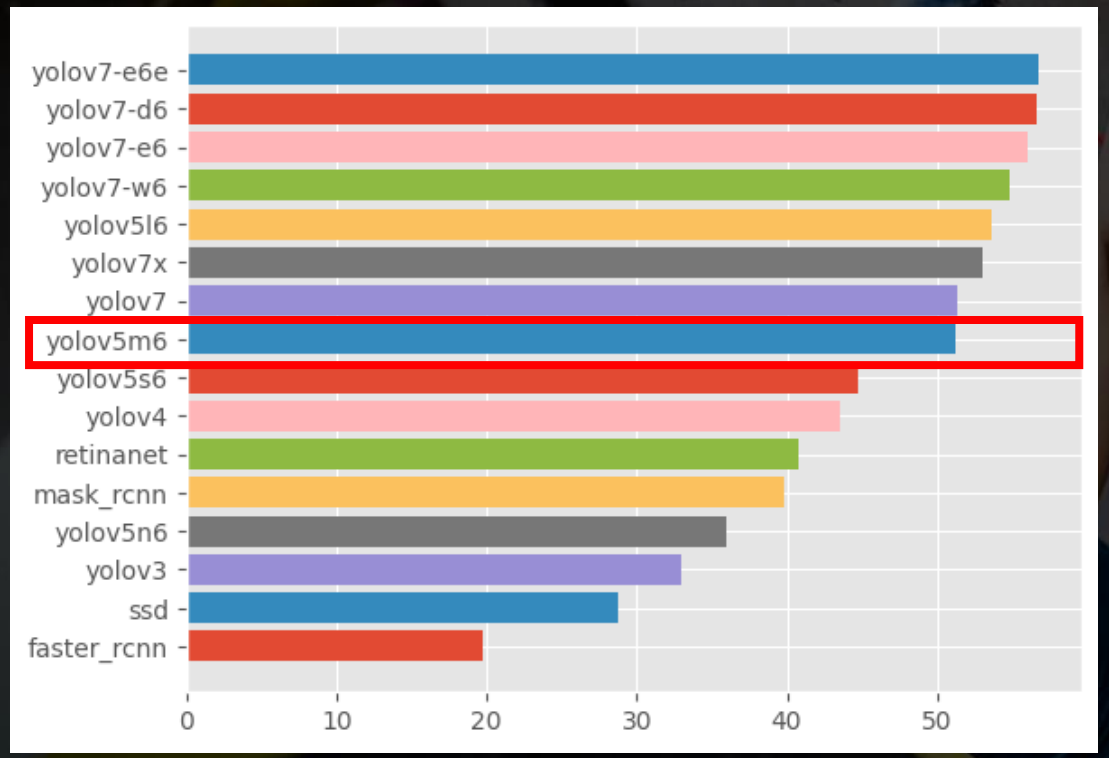
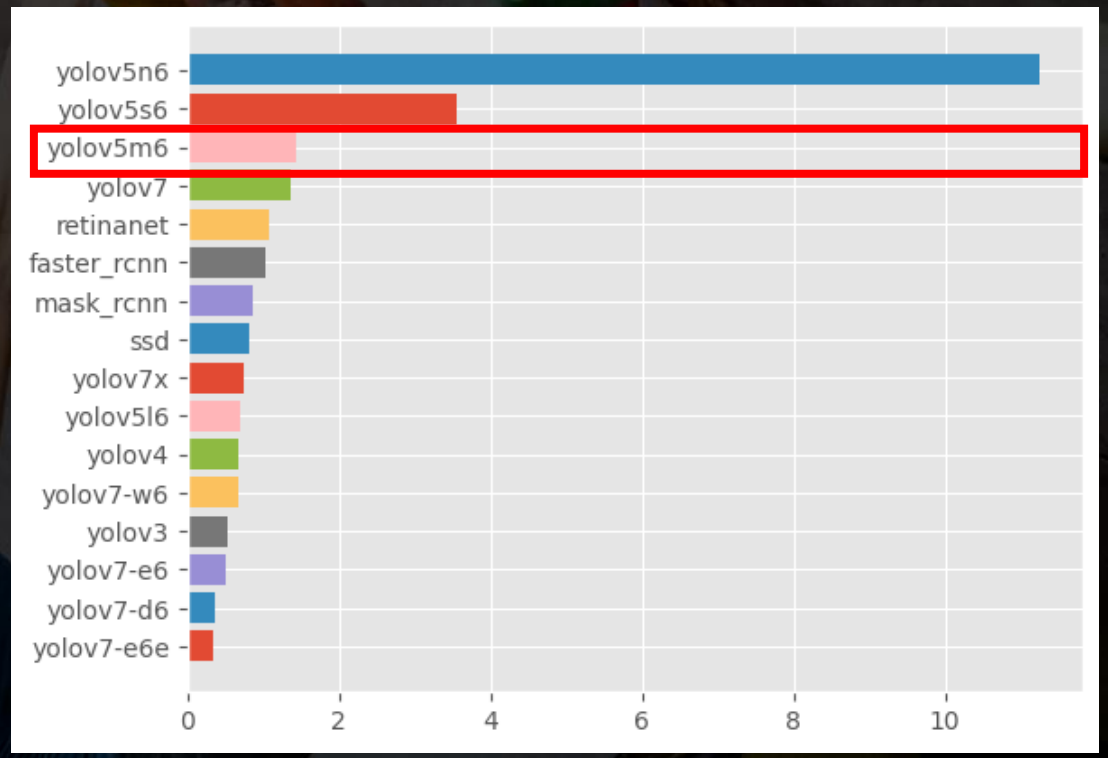
개발 진행사항 - 컴퓨터비전



mAP



Information Density

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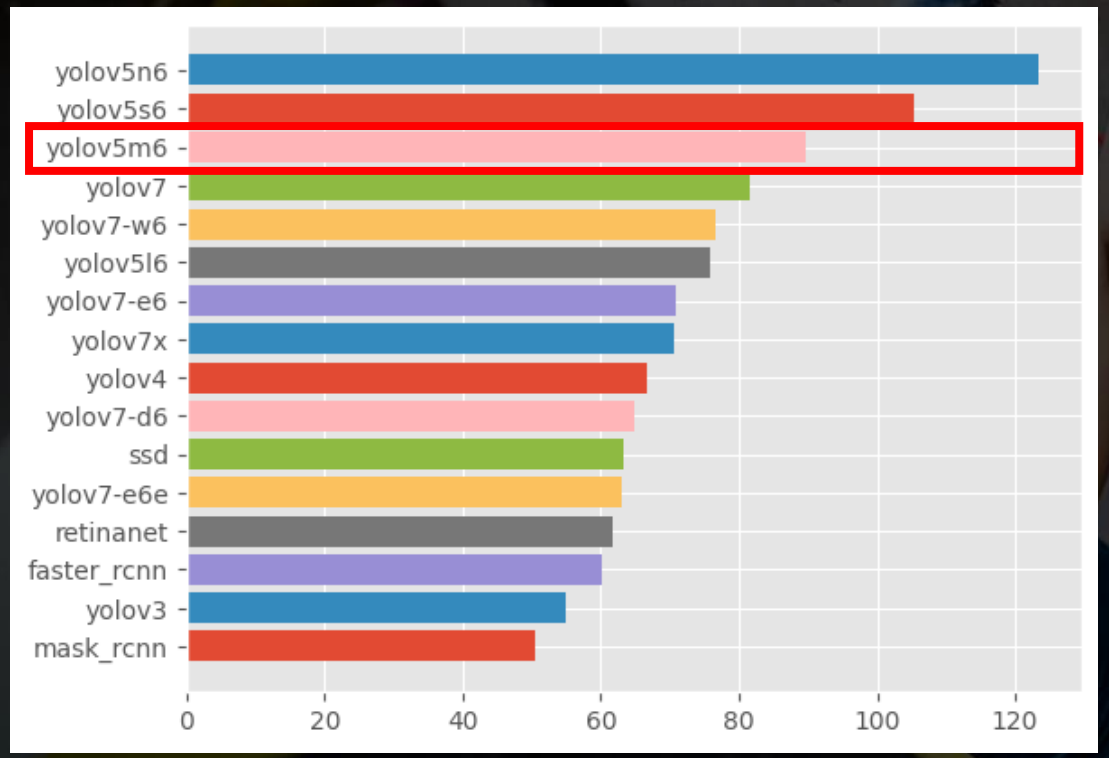
개발 진행사항 - 컴퓨터비전



NetScore

NetScore: Towards Universal Metrics for Large-scale Performance Analysis of Deep Neural Networks for Practical On-Device Edge Usage

Alexander Wong[1,2]
[1]Waterloo Artificial Intelligence Institute, University of Waterloo, Waterloo, ON, Canada
[2]DarwinAI Corp., Waterloo, ON, Canada
a28wong@uwaterloo.ca

$$\Omega(N) = 20log\left(\frac{a(N)^{\alpha}}{p(N)^{\beta}m(N)^{\gamma}}\right)$$

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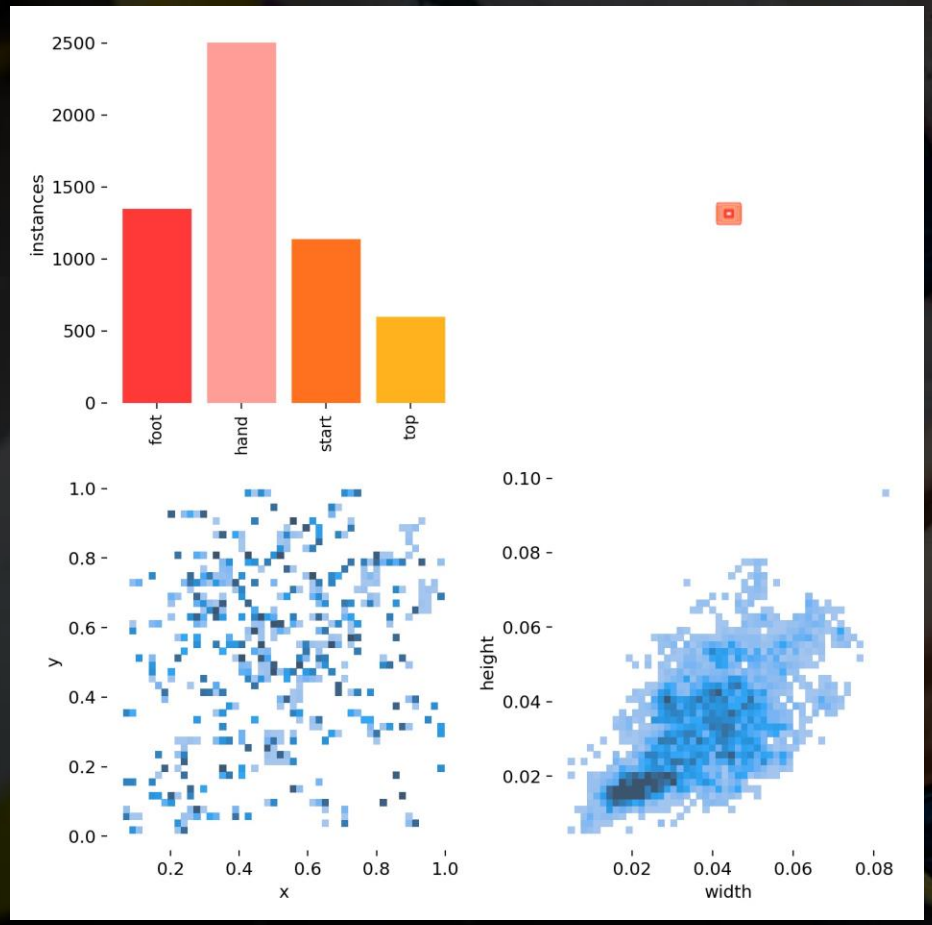
개발 진행사항 - 컴퓨터비전



- 데이터셋 제작 중
  - Annotation data - 600장
  - Train validation ratio - 9:1
- 문제점
  - Unbalanced label
  - Small object detection

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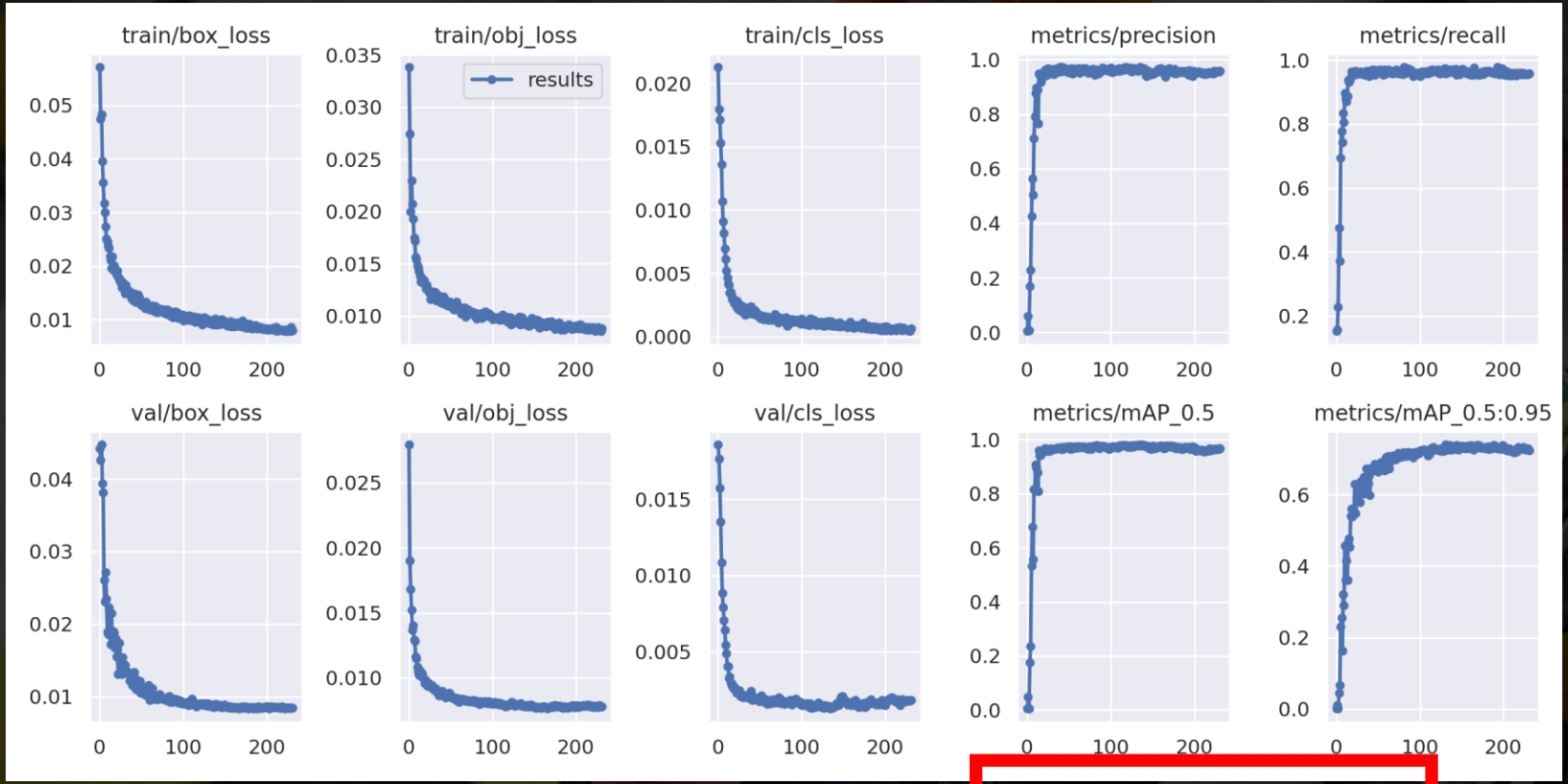
개발 진행사항 - 컴퓨터비전



낮은 precision

model	mAP_0.5	mAP 0.5:0.95	precision	recall
yolov5m6	0.9801	0.7399	0.9644	0.9576

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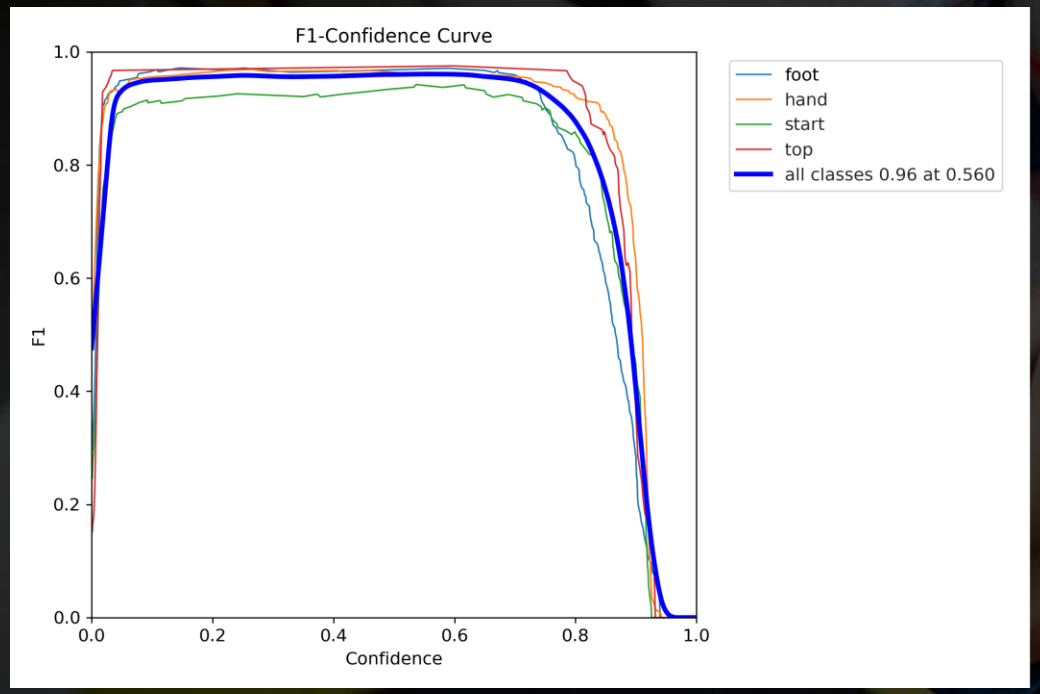
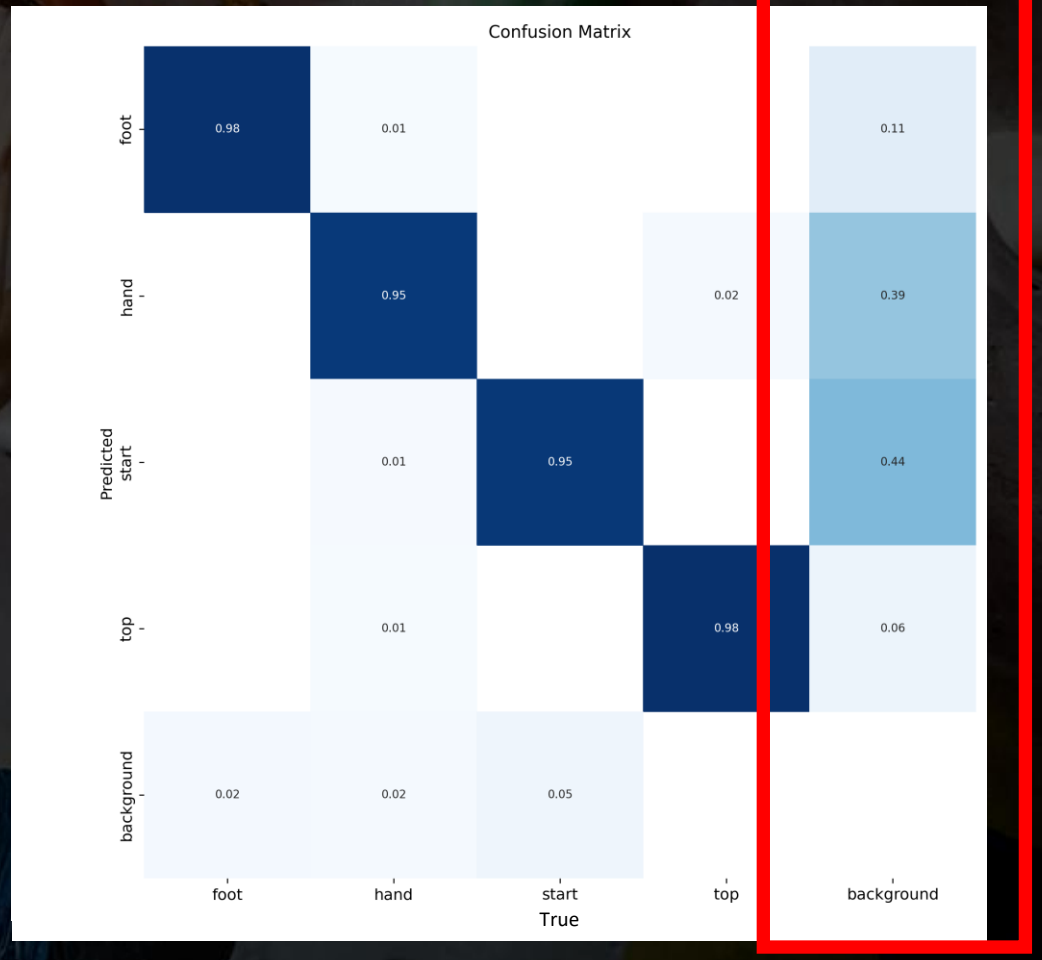
개발 진행사항 - 컴퓨터비전



낮은 Precision



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백엔드

- 각종 환경 세팅
  - 인프라
  - CI/CD
  - 모니터링

1

2

3

4

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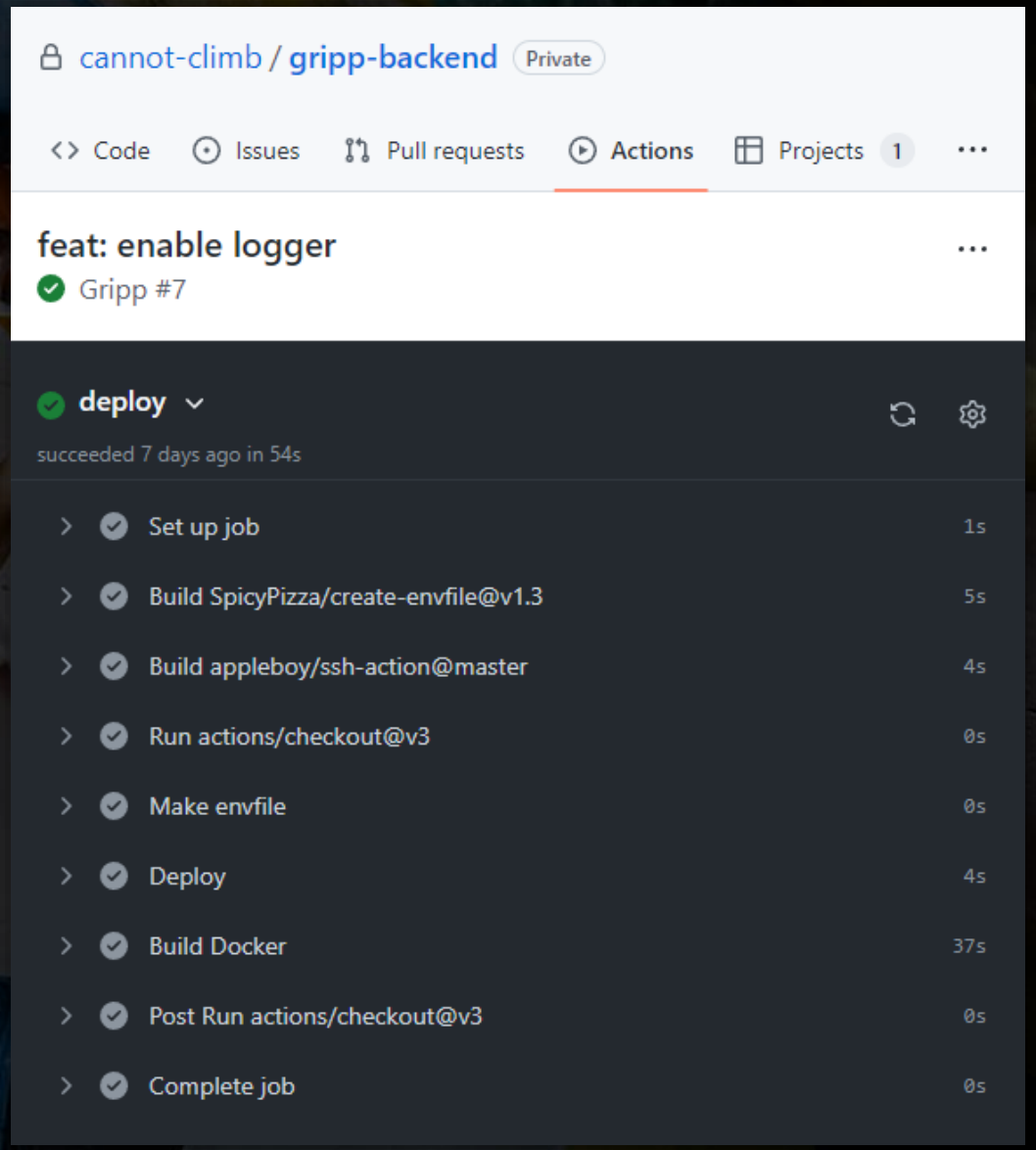
백엔드

- API 서버 24시간 구동 시작
  - https://gripp.dev.njw.kr
  - GitHub Actions을 이용하여 레포지토리에 push되는 내용을 실서버에 지속적으로 자동 배포

    (Continuous Deployment)



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백엔드

- API 서버 24시간 구동 시작
  - Docker를 이용한 컨테이너 기반 빌드/배포 (일관성, 이식성, 경량 가상 환경)

```
[+] Building 2.1s (14/14) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 32B
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load metadata for docker.io/library/eclipse-temurin:17
 => [gradle 1/7] FROM docker.io/library/eclipse-temurin:17@sha256:b87960fb0e8a55
 => [internal] load build context
 => => transferring context: 7.60kB
 => CACHED [stage-1 2/3] WORKDIR /gripp
 => CACHED [gradle 2/7] WORKDIR /gradle
 => CACHED [gradle 3/7] COPY gradlew build.gradle settings.gradle .
 => CACHED [gradle 4/7] COPY ./gradle ./gradle
 => CACHED [gradle 5/7] RUN ["./gradlew", "dependencies", "--no-daemon"]
 => CACHED [gradle 6/7] COPY . .
 => CACHED [gradle 7/7] RUN ["./gradlew", "clean", "build", "--no-daemon"]
 => CACHED [stage-1 3/3] COPY --from=gradle /gradle/build/libs/*.jar ./gripp.jar
 => exporting to image
 => => exporting layers
 => => writing image sha256:b2cb0b1dfe211a8116d4fc1053d2b1700423643d0a5a39da1657
 => => naming to docker.io/library/gripp-backend-gripp
[+] Running 1/0
 ⠿ Container gripp-backend-gripp-1  Running
```

```
ubuntu@njw1204-arm:~/gripp-backend$ docker compose logs
gripp-backend-gripp-1  |
gripp-backend-gripp-1  |   .   ____          _            __ _ _
gripp-backend-gripp-1  |  /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
gripp-backend-gripp-1  | ( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
gripp-backend-gripp-1  |  \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
gripp-backend-gripp-1  |   '  |____| .__|_| |_|_| |_\__, | / / / /
gripp-backend-gripp-1  |  =========|_|==============|___/=/_/_/_/
gripp-backend-gripp-1  |  :: Spring Boot ::                (v2.7.3)
gripp-backend-gripp-1  |
gripp-backend-gripp-1  | 2022-10-03 21:04:47.493  INFO 1 --- [           main] m.p.springdotenv.DotenvPr
opertySource    : Initializing Dotenv with DotenvConfig{directory=null, filename=null, ignoreIfMalformed
=false, ignoreIfMissing=false, systemProperties=false, prefix=null}
gripp-backend-gripp-1  | 2022-10-03 21:04:47.526  INFO 1 --- [           main] kr.njw.gripp.GrippApplica
tion            : Starting GrippApplication using Java 17.0.4.1 on aa51981887b8 with PID 1 (/gripp/gripp
.jar started by root in /gripp)
gripp-backend-gripp-1  | 2022-10-03 21:04:47.528  INFO 1 --- [           main] kr.njw.gripp.GrippApplica
tion            : No active profile set, falling back to 1 default profile: "default"
gripp-backend-gripp-1  | 2022-10-03 21:04:49.128  INFO 1 --- [           main] .s.d.r.c.RepositoryConfig
urationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
gripp-backend-gripp-1  | 2022-10-03 21:04:49.227  INFO 1 --- [           main] .s.d.r.c.RepositoryConfig
urationDelegate : Finished Spring Data repository scanning in 87 ms. Found 4 JPA repository interfaces.
gripp-backend-gripp-1  | 2022-10-03 21:04:50.114  INFO 1 --- [           main] o.s.b.w.embedded.tomcat.T
omcatWebServer  : Tomcat initialized with port(s): 8080 (http)
gripp-backend-gripp-1  | 2022-10-03 21:04:50.126  INFO 1 --- [           main] o.apache.catalina.core.St
andardService   : Starting service [Tomcat]
gripp-backend-gripp-1  | 2022-10-03 21:04:50.127  INFO 1 --- [           main] org.apache.catalina.core.
StandardEngine  : Starting Servlet engine: [Apache Tomcat/9.0.65]
gripp-backend-gripp-1  | 2022-10-03 21:04:50.207  INFO 1 --- [           main] o.a.c.c.C.[Tomcat].[local
host].[/]       : Initializing Spring embedded WebApplicationContext
gripp-backend-gripp-1  | 2022-10-03 21:04:50.208  INFO 1 --- [           main] w.s.c.ServletWebServerApp
licationContext : Root WebApplicationContext: initialization completed in 2610 ms
gripp-backend-gripp-1  | 2022-10-03 21:04:50.596  INFO 1 --- [           main] o.hibernate.jpa.internal.
util.LogHelper  : HHH000204: Processing PersistenceUnitInfo [name: default]
gripp-backend-gripp-1  | 2022-10-03 21:04:50.670  INFO 1 --- [           main] org.hibernate.Version
                : HHH000412: Hibernate ORM core version 5.6.10.Final
gripp-backend-gripp-1  | 2022-10-03 21:04:50.913  INFO 1 --- [           main] o.hibernate.annotations.c
ommon.Version   : HCANN000001: Hibernate Commons Annotations {5.1.2.Final}
gripp-backend-gripp-1  | 2022-10-03 21:04:51.034  INFO 1 --- [           main] com.zaxxer.hikari.HikariD
ataSource       : HikariPool-1 - Starting...
gripp-backend-gripp-1  | 2022-10-03 21:04:51.389  INFO 1 --- [           main] com.zaxxer.hikari.HikariD
ataSource       : HikariPool-1 - Start completed.
gripp-backend-gripp-1  | 2022-10-03 21:04:51.438  INFO 1 --- [           main] org.hibernate.dialect.Dia
lect            : HHH000400: Using dialect: org.hibernate.dialect.MySQL8Dialect
gripp-backend-gripp-1  | 2022-10-03 21:04:52.310  INFO 1 --- [           main] o.h.e.t.j.p.i.JtaPlatform
Initiator       : HHH000490: Using JtaPlatform implementation: [org.hibernate.engine.transaction.jta.pla
tform.internal.NoJtaPlatform]
```

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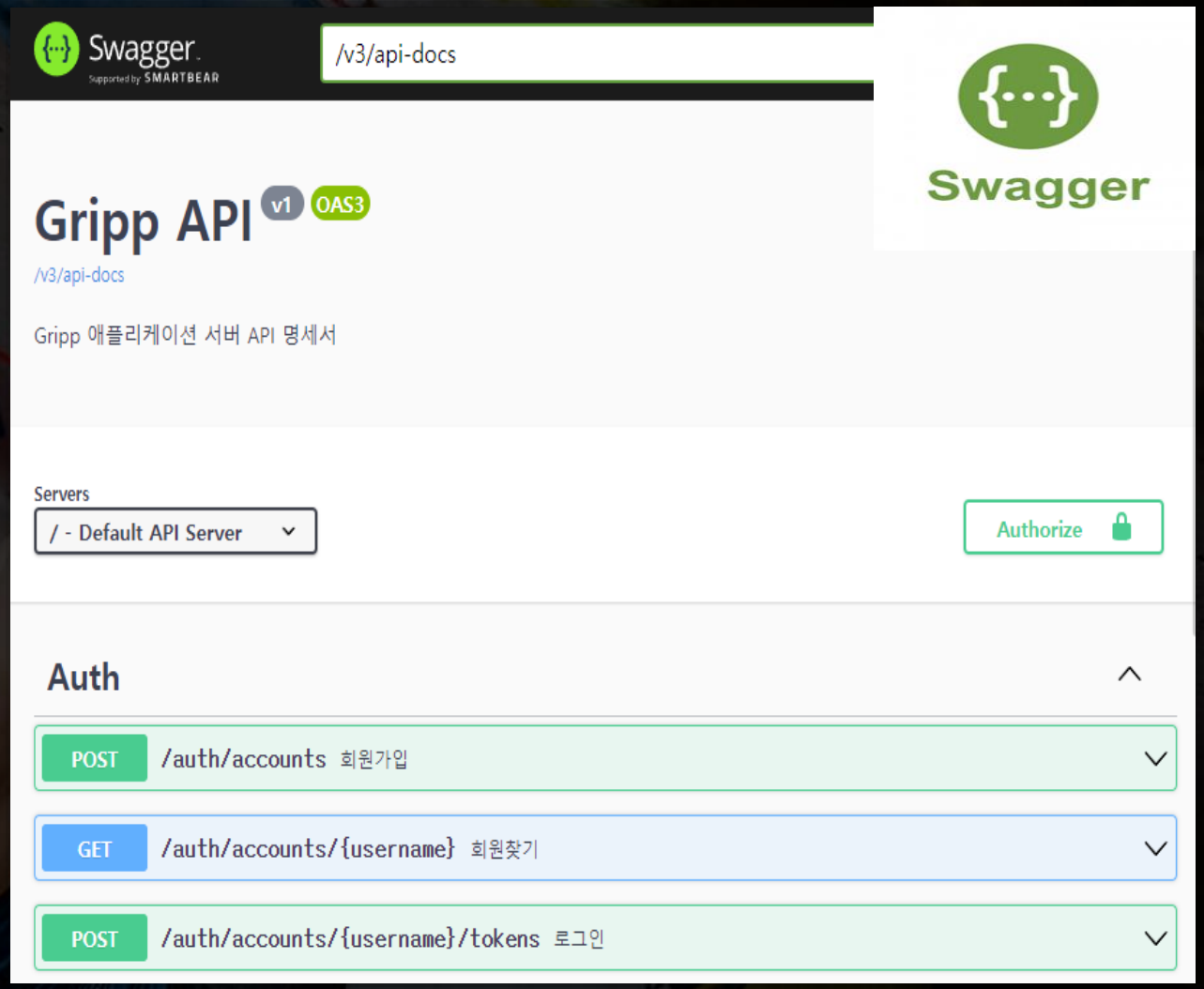
백엔드

- API 문서화
  - API 초안은 Notion으로 작성 및 공유
  - Swagger를 통한 코드 기반 API 문서 자동화



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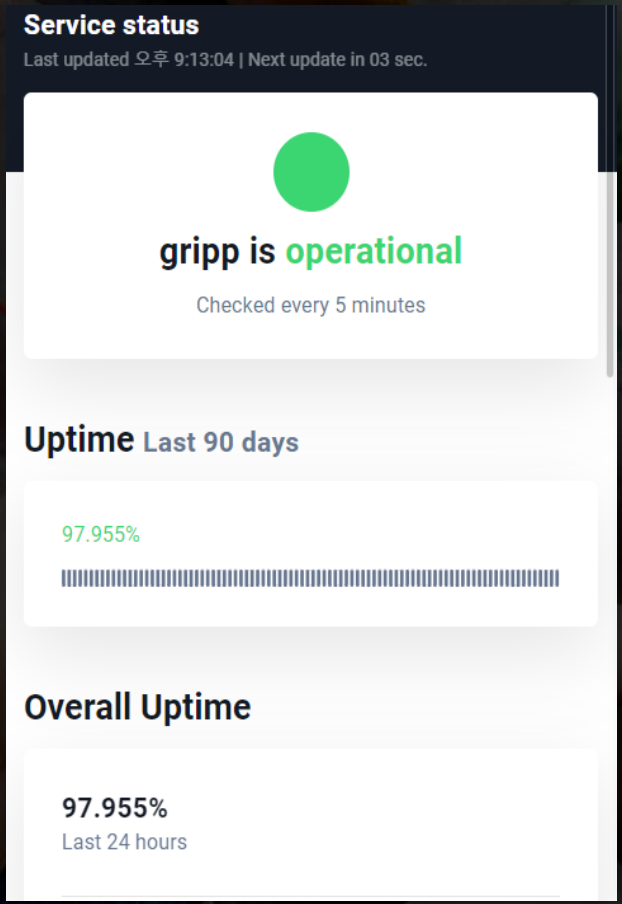
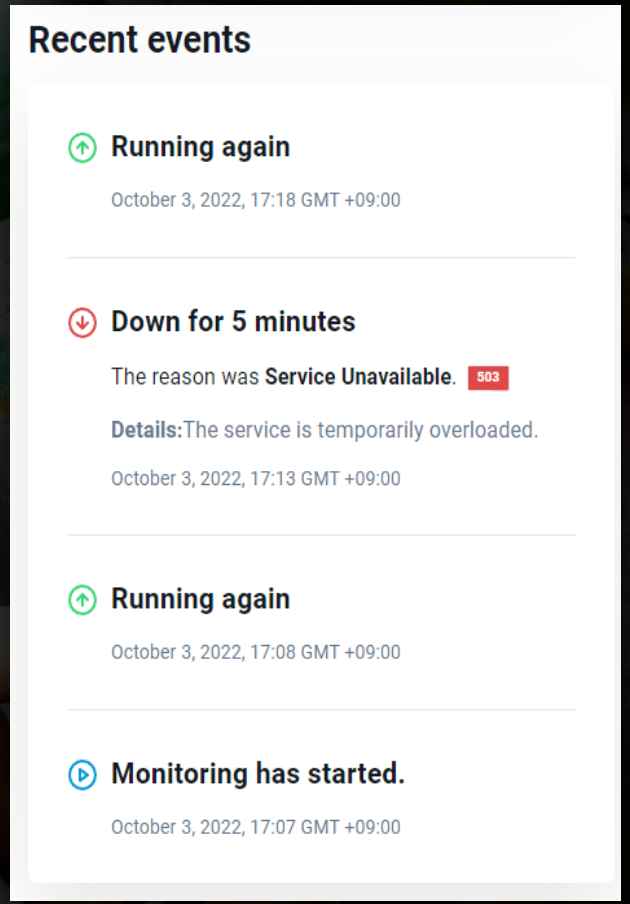
백엔드

- API 모니터링
  - Spring Actuator를 통한 서버 health 체크
  - cpu, disk, network 등 주요 metrics 모니터링
  - 서버 다운 시 이메일 알림





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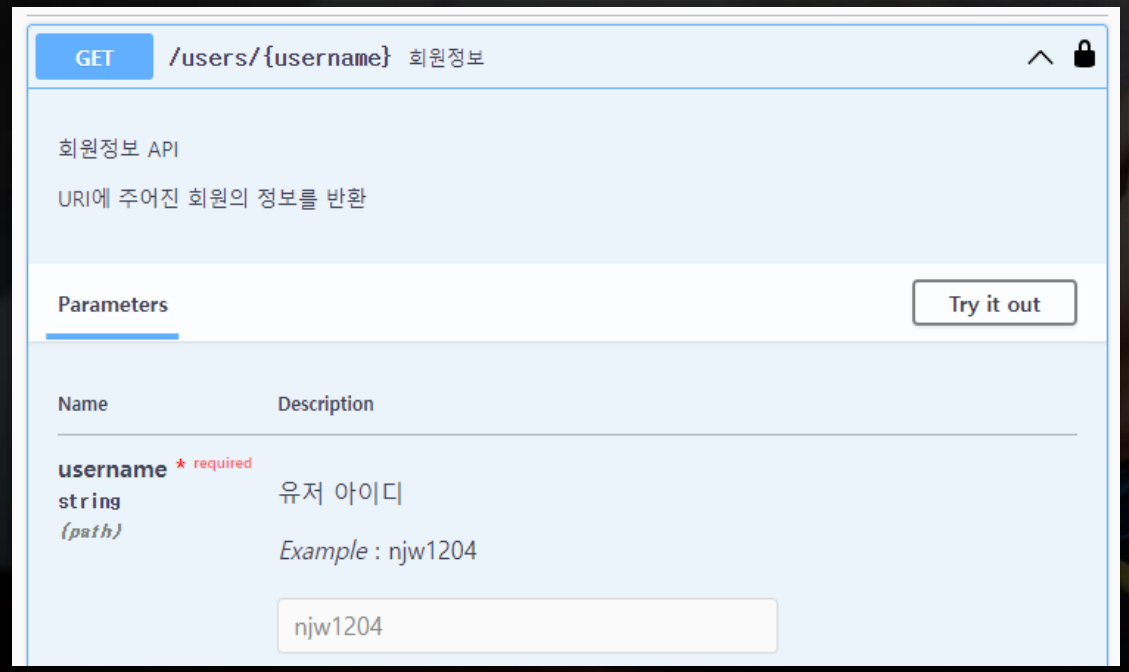
백엔드

- Spring Security 기반 JWT(JSON Web Token) 인증 구현 완료
- 회원 시스템과 관련된 API 구현 중



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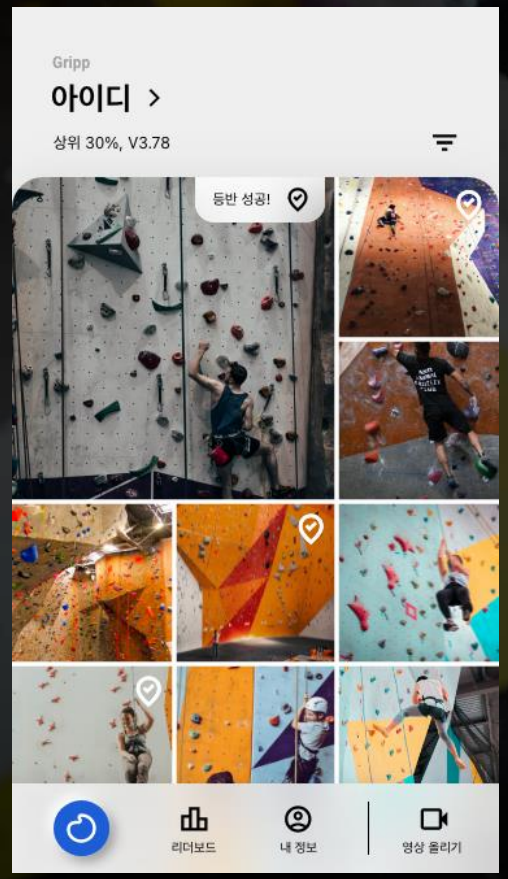
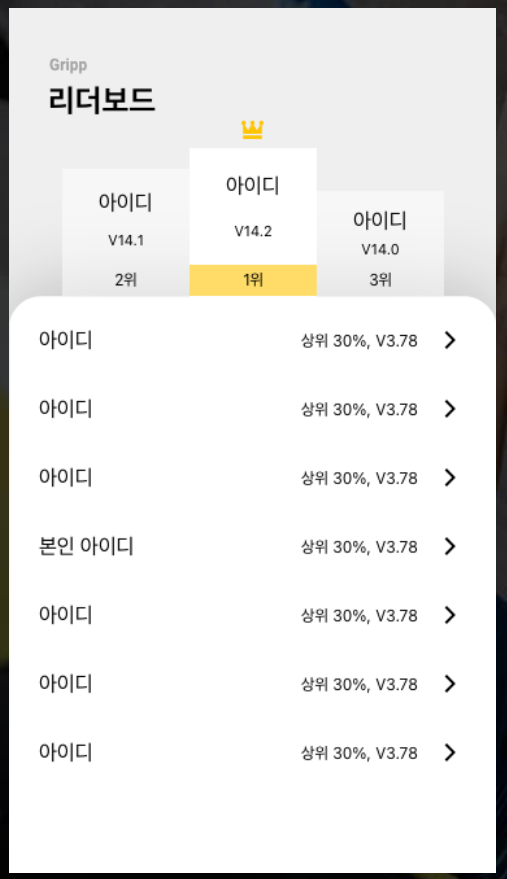
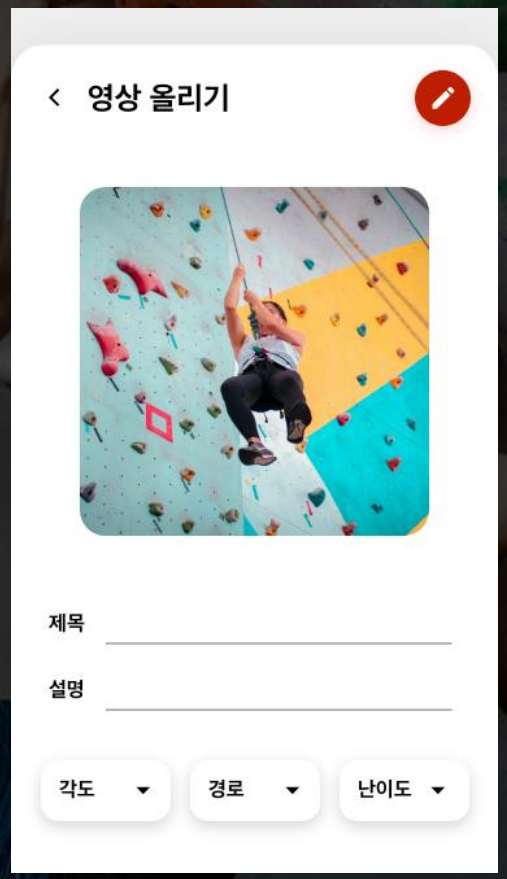
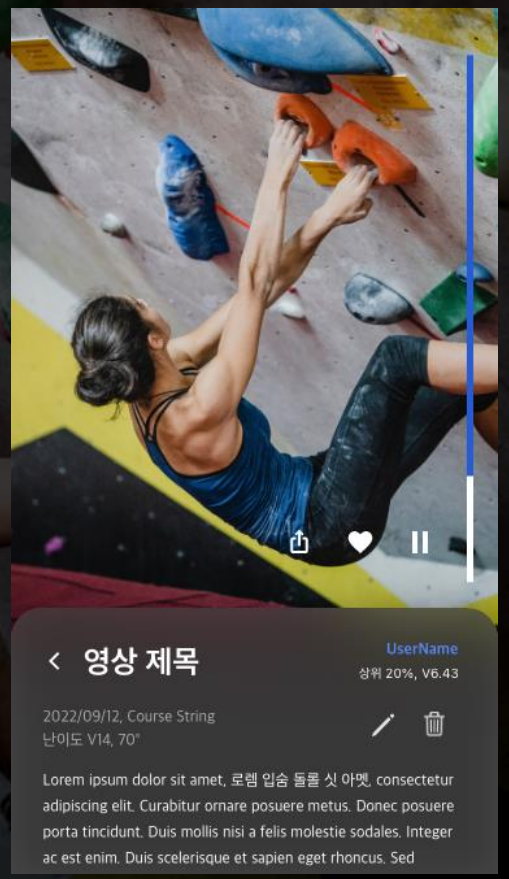
디자인


Gripp
아이디 >
상위 30%, V3.78
등반 성공!
리더보드
내 정보
영상 올리기


Gripp
리더보드
아이디
V14.1
2위
아이디
V14.2
1위
아이디
V14.0
3위
아이디 상위 30%, V3.78
아이디 상위 30%, V3.78
아이디 상위 30%, V3.78
본인 아이디 상위 30%, V3.78
아이디 상위 30%, V3.78
아이디 상위 30%, V3.78
아이디 상위 30%, V3.78


영상 올리기
제목
설명
각도
경로
난이도


영상 제목
UserName
상위 20%, V6.43
2022/09/12, Course String
난이도 V14, 70°
Lorem ipsum dolor sit amet, 로렘 입숨 돌롤 싯 아멧, consectetur adipiscing elit. Curabitur ornare posuere metus. Donec posuere porta tincidunt. Duis mollis nisi a felis molestie sodales. Integer ac est enim. Duis scelerisque et sapien eget rhoncus. Sed

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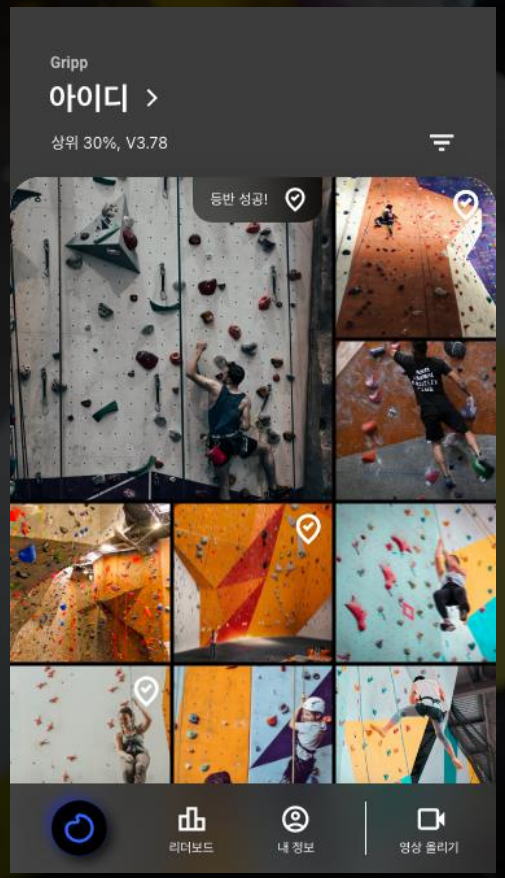
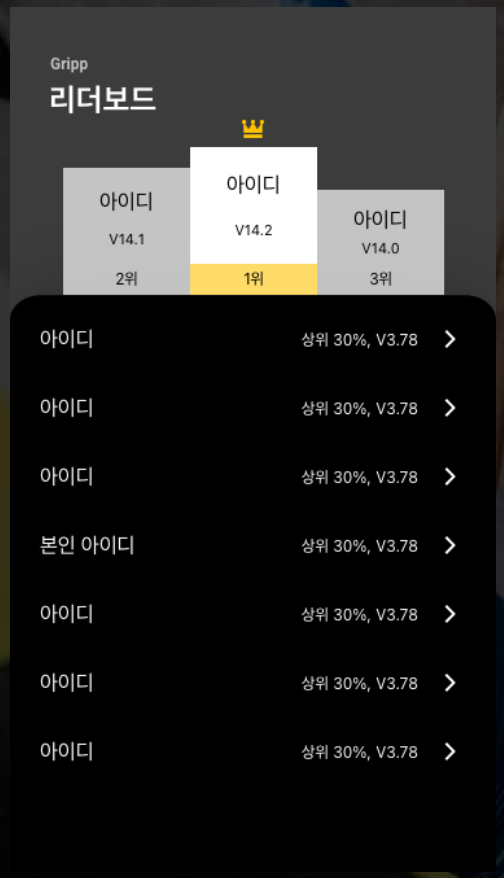
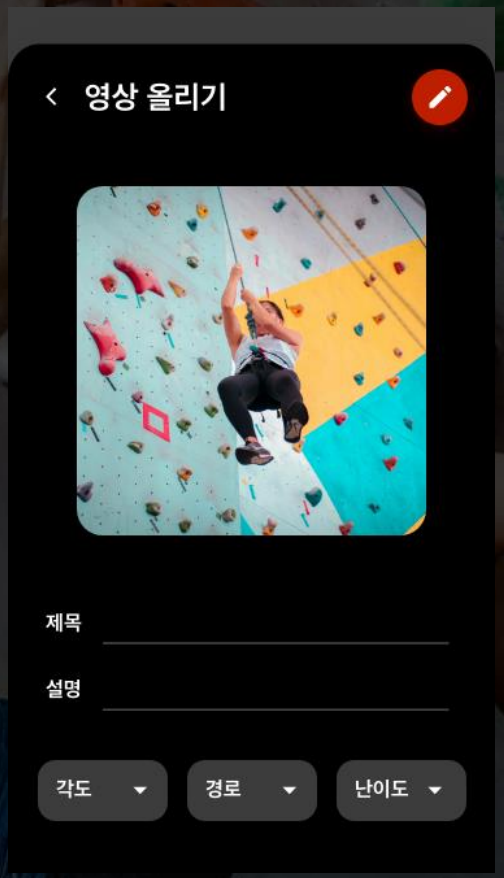
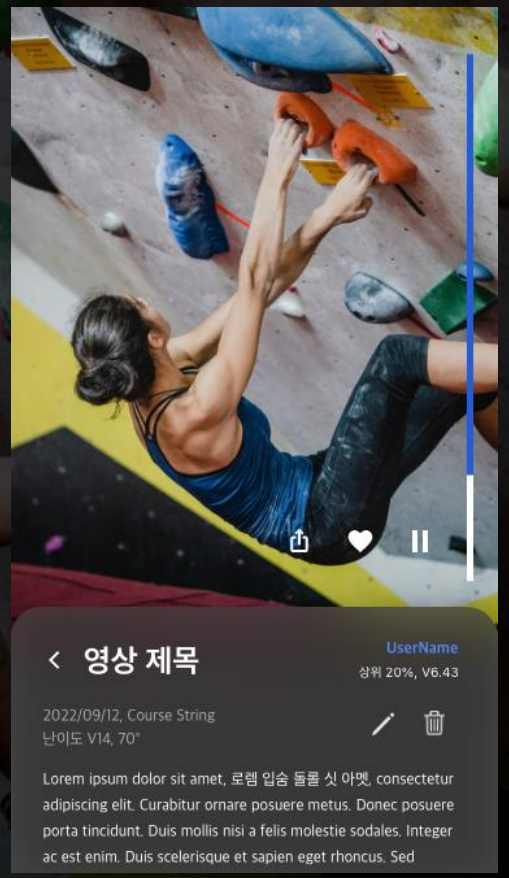
앱

최종 디자인 + 다크 모드









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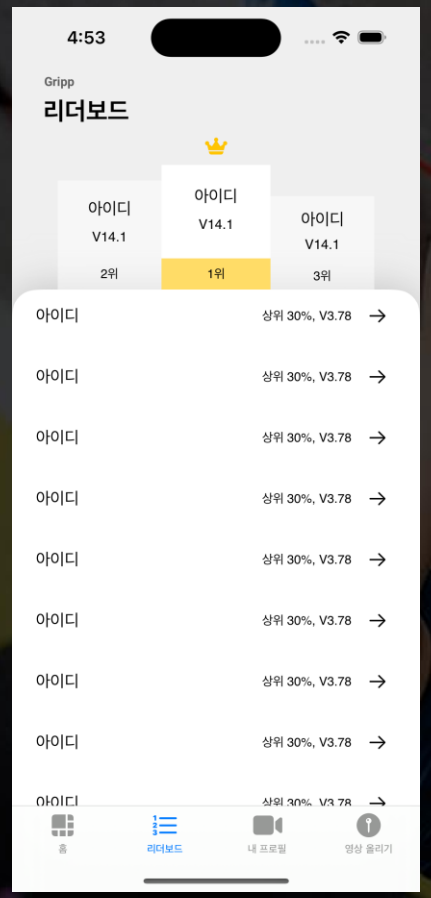
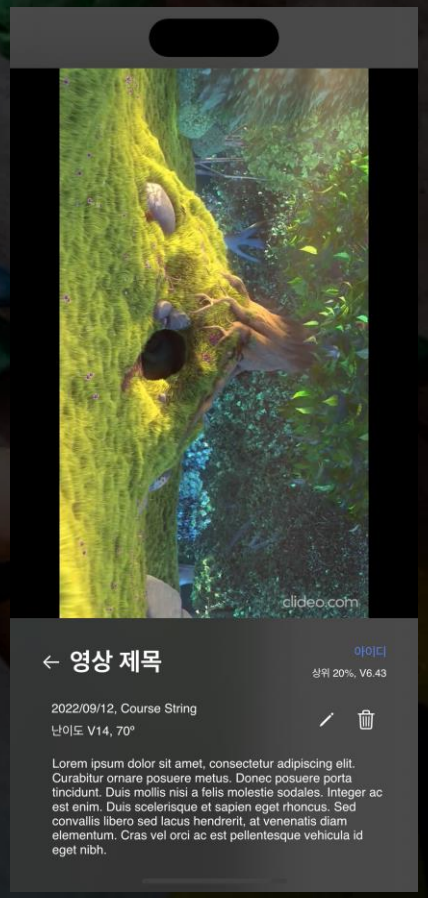
Swift 5.7, Xcode 14, SwiftUI 4





...

HLS(.m3u8) video playback

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진행사항 요약

- 아이디어 논의
- 기획 확정
- 기술 스택 학습
- 전반적인 구현 과정 논의

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진행사항 요약

- 컴퓨터비전

  데이터셋 수집 및 기준모델 학습

- 백엔드

  각종 환경 세팅, API 서버 구동 시작, 회원 API 구현 중

- 앱

  화면 구성 완료. 이벤트 처리 중, 서버 통신 베이스 설계 중

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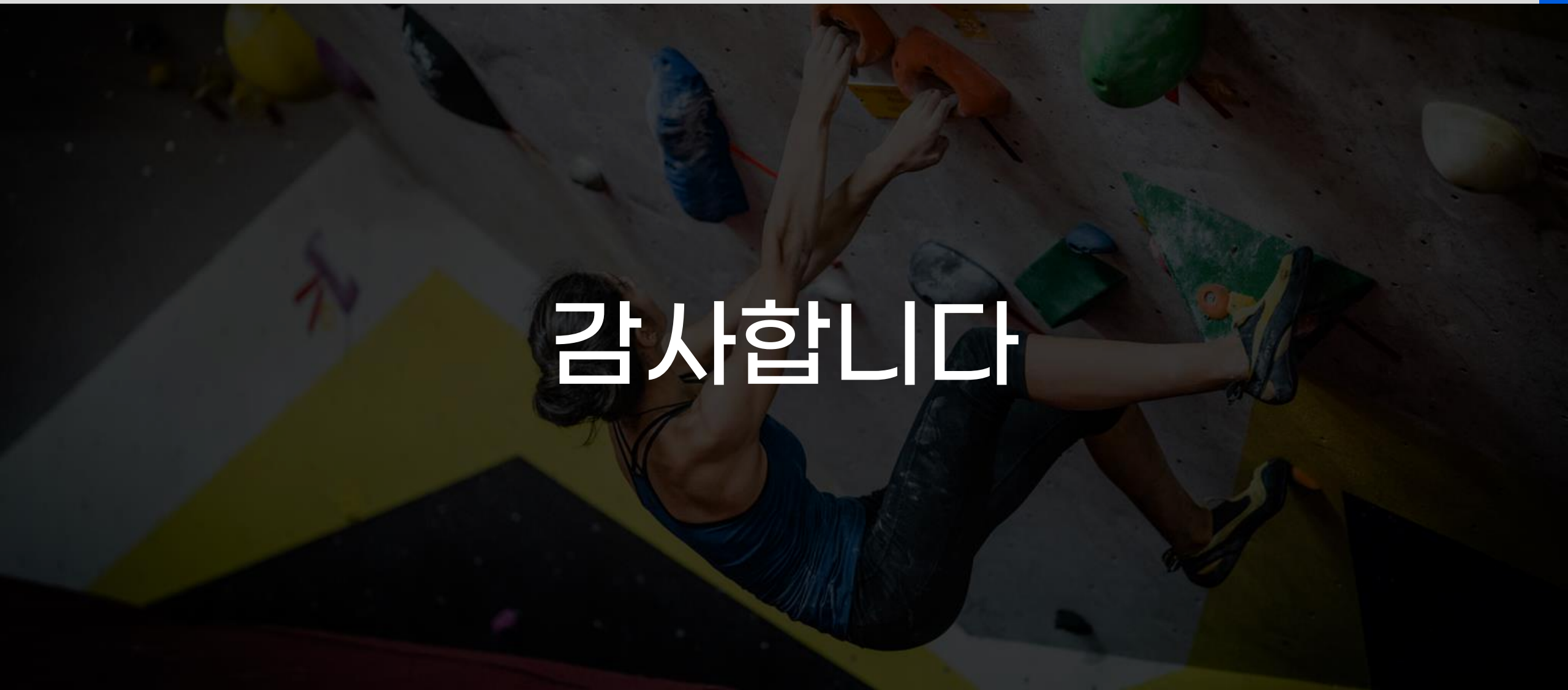

감사합니다